Poems Syndrome and Disease Produced by Other Monoclonal IGs

83

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83.1 POEMS Syndrome

83.1.1 Introduction

POEMS syndrome (acronym of: *p*olyradiculo-neuropathy, *o*rganomegaly, *e*ndocrinopathies, *m*onoclonal protein and dermopathy/*s*kin) is a rare multisystemic disease due to an underlying plasma cell neoplasm. The pathogenesis of the syndrome is not well understood. Other names of the POEMS syndrome that are less frequently used are osteosclerotic myeloma, Takatsuki syndrome, or Crow-Fukase syndrome.

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83.1.2 Clinical and Laboratory Manifestations

• Male predominance • Age (maximun incidence): 50–60 years

Polyneuropathy	Typically demyelinating. Peripheral, ascending, symmetrical, and affecting both sensation and motor function. It is the dominant characteristic
Organomegaly	Hepatomegaly (50%), splenomegaly, or lymphadenopathy
Endocrinopathy	Present in 84%: gonadal, thyroid, pituitary, parathyroid, pancreatic, adrenal (in order of frequency, and many times multiple)
Monoclonal protein	Almost always λ light chain. Usually Ig A or IgG and $\leq$ 3 g/dL. Bone marrow smear: <5–10% plasma cells
Skin changes	Hyperpigmentation, hypertrichosis, glomeruloid hemangiomata, white nails, plethora, acrocyanosis, flushing
Other important manifestations	– Sclerotic bone lesions[a] (95%) – Castleman disease (in 11–30%) – Papilledema (in one-third of patients) – Extravascular volume overload – Thrombocytosis (in 54%) – VEGF[b] elevation

[a]Radiology and CT/PET can be useful
[b]*VEGF Vascular endothelial growth factor* is the cytokine that correlates best with disease activity. The helpful cut-off for plasma and serum VEGF levels for diagnosis are ≥200 pg/mL (specificity 95%, sensitivity 68%) and ≥ 1920 pg/mL (specificity 98%, sensibility 73%), respectively

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83.1.3 Diagnosis

Not all the features within the acronym are required to make the diagnosis. There are other relevant features not included in the POEMS acronym also important: PEST (*p*apilledema, *e*xtravascular volume overload, *s*clerotic bone lesions, *t*hrombocytosis/erythrocytosis), elevated VEGF levels, abnormal pulmonary function tests, and a predisposition to thrombosis. There is a Castleman variant of POEMS syndrome that may be associated with a clonal plasma cell disorder. When Castleman disease variant of POEMS syndrome occurs without evidence of plasma cell disorder, then this entity should be considered separately.

83.1.3.1 Criteria for the Diagnosis of POEMS Syndrome

The diagnosis of POEMS syndrome is confirmed when:

- Both mandatory major criteria +.
- another of the other 3 major criteria +,
- at least one of the minor criteria.

Mandatory major criteria
Polyneuropathy Clonal plasma cell dyscrasia (monoclonal immunoglobulin)
Other major criteria
Castleman disease Sclerotic bone lesions VEGF elevated
Minor criteria
Organomegaly (splenomegaly, hepatomegaly, or lymphadenopathy) Extravascular volume overload (edema, pleural effusion, or ascites) Endocrinopathy[a] Skin changes Papilledema Thrombocytosis/erythrocytosis[b]
Other symptoms and signs
Digital clubbing Weight loss Hyperhidrosis Low vitamin B_{12} values Diarrhoea Pulmonary hypertension/restrictive lung disease Thrombosis

Adapted from Dispenzieri A, Am J Hematol 2017

[a]Hypogonadism is the most frequent and because of the high prevalence of diabetes mellitus and thyroid abnormalities, these two last abnormalities alone are not sufficient to meet this minor criterion

[b]Anemia and/or thrombocytopenia are rare, unless associated with Castleman disease

83.1.4 Prognosis

- Chronic course, median survival of nearly 14 years, rarely progression to multiple myeloma.
- The number of POEMS features does not affect survival.
- Risk factors associated to better survival → albumin > 3.2 g/dL, achievement of a complete hematological response and younger age. Lower VEGF levels, better response to treatment.
- Risk factors associated to shorter survival → clubbing, extravascular volume overload, respiratory symptoms, papilledema, and coexisting Castleman disease.
- Thrombocytosis and high bone marrow infiltration are associated with risk of cerebrovascular accidents.
- Patients candidates for radiation therapy have a better overall survival.

83.1.5 Standard treatment

Need to differentiate between POEMS with chronic inflammatory demyelinating polyneuropathy and CANOMAD (chronic ataxic neuropathy with opthalmoplegia, M-protein, cold agglutinins, and disialosyl antibodies).

- **In case of an isolated bone lesion** (or multiple, but localized):
  Radiotherapy (e.g., 35–50 Gy) to affected site(s) → improve the symptoms of POEMS syndrome and can be curative
- **Rest of patients (disseminated disease)**:
  - → Lenalidomide + dexamethasone is mainstay of frontline therapy

- → Alternatives include melphalan + dexamethasone, thalidomide + dexamethasone, bortezomib + dexamethasone (these last two agents are of limited use due to the intrinsic risk of peripheral neuropathy), cyclophosphamide ± dexamethasone
- → Plasmapheresis, IgIV, IFN-α, tamoxifen, trans-retinoic acid, bevacizumab (anti-VEGF agent), argatroban, and strontium-89 (mostly single case reports)
- → Attention to supportive care is mandatory (physical therapy, orthotics, etc.)
- → Autologous HCT (see Sect. 83.1.6)

83.1.5.1 Response Criteria

Monitoring the response to treatment in POEMS syndrome is a challenge. Patients must be followed carefully comparing the deficits to baseline. VEGF is an imperfect marker due to discordances between disease activity and response. The size of monoclonal protein is typically small making standard multiple myeloma response criteria inapplicable. Patients can present clinical benefit without M-protein response therefore a clinical scoring system which can focus on organ-specific response would be useful clinically. So, response criteria for POEMS syndrome could be done as follows: hematological response using a modified amyloid response criteria; VEGF response; CT/PET response; and a simplified organ response (polyneuropathy assessment, pulmonary function tests, and extravascular overload).

83.1.6 Autologous Hematopoietic Cell Transplantation (Autologous HCT)

Background	– In multiple myeloma → autologous HCT → high rate and depth of responses – In amyloidosis, a disease with similarities to POEMS syndrome with "low tumor" burden → autologous HCT → long remissions obtained
Indication	POEMS syndrome with disseminated disease: • Good general condition • Either upfront or after suitable induction therapy (lenalidomide + dexamethasone), though mostly after induction to control symptoms prior to autologous HCT
Conditioning	Melphalan 140–200 mg/m^2
Stem cell source[a,b]	PBSC, mobilization with G-CSF ± Cy (1.5–3 g/m^2). Mobilize early if lenalidomide+dexamethasone used.
Morbidity	High rate of engraftment syndrome (up to 50%) (see Chap. 20), important to recognize and treat promptly with prednisone. In these cases, higher than expected transfusion need and delayed engraftment. No organ toxicities as observed in amyloidosis
Mortality[c]	As in other autologous HCT, recently reported 3.3% 1-year NRM[d]
Response[c,d]	Hematological CR can be achieved by 3 months post-autologous HCT, but neurologic response is usually delayed to 6–9 months, but full recovery can take up to 2–3 years. PET/CT evidence of response can take upto 12 months

[a]Mobilization failure is described, for this reason, if there is no response after 3 courses of induction, proceed to mobilization

[b]The incidence of engraftment syndrome can be reduced if mobilization is done with cyclophosphamide + G-CSF.

[c]Cook G, Hematologica 2017

[d]In the largest series, 3-y PFS 84% and OS 94%, and 5-y PFS 74% and OS 89%

83.2 Monoclonal Ig Deposition Disease

83.2.1 Introduction

Monoclonal Ig deposition is a clonal plasma cell discrasia in which light-chain and/or heavy-chain subunits of Igs form nonfibrillar deposits in various tissues, causing organ dysfunction. Light-chain deposition disease is the most common of these entities.

83.2.2 Clinical Manifestation/ Laboratory

Kidney	Always affected. Immunofluorescence shows deposition of light chains along glomerular and tubular basement membranes → nodular glomerulosclerosis. Deposits are nonfibrillar, almost always composed κ chain and do not stain with Congo red dye → nephrotic syndrome, hypertension, and rapidly progressing renal insufficiency
Heart and liver	Less frequent affected. Restrictive cardiopathy, myocardial infarction, cholestatic jaundice, hepatic failure
Monoclonal gammopathy	Electrophoresis, immunofixation of serum and/or urine, serum-free light chain measurement

83.2.3 Diagnosis

Biopsy of the affected organ (almost always kidney).

83.2.4 Treatment

- Controversial, not standard due to the low incidence. Conventional chemotherapy commonly used for MM is unsatisfactory.
  - Melphalan + Prednisone
  - VAD
  - Thalidomide ± DXM, bortezomib + DXM
  - Autologous HCT (see Sect. 83.2.5)

83.2.5 Autologous Hematopoietic Cell Transplantation (Autologous HCT)

Background	– As in POEMS syndrome (see Sect. 83.1.6)
Indication	– Patients in good general condition and with basic requirements for autologous HCT – Patients not responding to previous MM-like treatment
Conditioning	Melphalan 140–200 mg/m^2
Stem cell source	PB, mobilization with G-CSF ± Cy (3 g/m^2)
Morbidity	Some patients require hemodialysis (HD) before and during the procedure. In that case, melphalan should be administered after HD
Mortality	As in other autologous HCT
Response	In the few cases reported: – Hematological responses are described secondary to the control of the monoclonal gammopathy. – It can improve renal function. In selected cases, kidney transplantation could be an option if the patient achieve a CR and remain in HD.

Further Reading

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